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A Comparative Machine Learning Framework for Predicting Childhood Pneumonia Prevalence: Integrating Household Crowding, Vaccination Status, and Passive Smoking as Socio-Environmental Determinants

--Manuscript Draft--

Manuscript Number:	PONE-D-26-48143
Article Type:	Research Article
Full Title:	A Comparative Machine Learning Framework for Predicting Childhood Pneumonia Prevalence: Integrating Household Crowding, Vaccination Status, and Passive Smoking as Socio-Environmental Determinants
Short Title:	A Comparative Machine Learning Framework for Predicting Childhood Pneumonia Prevalence
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Keywords:	Childhoodpneumonia,Machinelearning,Featureablation,CommunityExposureRiskScore ,SMOTE, SHAP, Probability calibration, Decision curve analysis, DeLong test, Fairness analysis, Household crowding, Vaccination, Low- and middle-income countries, Community health
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August 20, 2026

The Editors, PLOS ONE

Dear Editors,

We are pleased to submit our original research article, “A comparative machine learning framework for predicting childhood pneumonia prevalence: integrating household crowding, vaccination status, and passive smoking as socio-environmental determinants,” for consideration as a Research Article in PLOS ONE.

Childhood pneumonia remains the leading infectious cause of death in children under five, with over 95% of cases concentrated in low- and middle-income countries (LMICs). Machine learning has increasingly been applied to pediatric pneumonia risk prediction, but prior studies have generally reported discriminative performance (AUC) in isolation, without the feature ablation studies, probability calibration, decision curve analysis, fairness evaluation, and interpretability analyses increasingly expected of clinically trustworthy medical AI.

We present the Community-Aware Explainable Pneumonia Prediction Framework (CAEPPF), developed and validated on 799 pediatric patients from a tertiary hospital in Rabat, Morocco, using a publicly available, fully de-identified dataset (Mendeley Data, CC BY 4.0). CAEPPF integrates nine methodological components not previously reported together in this literature: a feature ablation study quantifying the incremental contribution of socio-environmental determinants, a data-driven Community Exposure Risk Score, benchmarking of eleven algorithms under a leakage-free SMOTE pipeline, formal statistical model comparison (DeLong/McNemar tests), probability calibration analysis, decision curve analysis, subgroup fairness evaluation, expanded SHAP interpretability, a ten-seed sensitivity analysis, and a deployable Community Pneumonia Screening Score intended for use by community health workers without laboratory infrastructure.

We believe this work suits PLOS ONE's broad scope and emphasis on methodological rigor and reproducibility, bridging predictive performance with the clinical-validity and deployability requirements essential for translating ML research into real-world community screening tools in resource-constrained settings. This is a secondary, fully de-identified data analysis; institutional review board approval was not required, as detailed in the Methods.

This is a Research Article. The manuscript is original, has not been published previously, and is not under consideration elsewhere, in whole or in part. All authors have approved the manuscript and agree with its submission to PLOS ONE. We have had no prior interactions with PLOS regarding this manuscript and have no opposed reviewers to declare.

Given the manuscript's focus on applied machine learning for pediatric infectious disease risk prediction in a global-health context, we suggest Academic Editors with expertise in clinical machine learning, pediatric infectious disease epidemiology, or global/public health informatics.

Thank you for considering our manuscript. We look forward to your response.

Sincerely,

Bishal Paul (Corresponding Author)

On behalf of all authors: Md Artad Chowdhury, Bishal Paul, Shafiul Alam Faysal, Md. Asif Mredha

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A comparative machine learning framework for predicting childhood pneumonia prevalence: integrating household crowding, vaccination status, and passive smoking as socio-environmental determinants

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Abstract

Childhood pneumonia is the leading infectious cause of mortality in children under five years of age, responsible for approximately 700,000 deaths annually, with over 95% of cases occurring in low- and middle-income countries (LMICs). This paper presents the Community-Aware Explainable Pneumonia Prediction Framework (CAEPPF), a comprehensive, clinically grounded machine learning (ML) framework developed and validated on 799 pediatric patients admitted to a tertiary hospital in Rabat, Morocco. CAEPPF integrates nine methodological components that, to our knowledge, have not previously been reported together in the pediatric pneumonia ML literature: (1) a feature ablation study quantifying the incremental discriminative contribution of socio-environmental variables (+1.8 AUC points, 0.861 to 0.879); (2) data-driven optimization of a novel Community Exposure Risk Score (CERS) using logistic regression coefficients; (3) benchmarking of eleven supervised ML algorithms under a leakage-free Synthetic Minority Oversampling Technique (SMOTE) pipeline with

hyperparameter tuning, achieving a Random Forest cross-validated AUC of 0.887 (95% CI: 0.825–0.949); (4) DeLong and McNemar statistical comparison tests across all top-model pairs; (5) full probability calibration analysis (Brier Score, Expected Calibration Error) identifying Random Forest as the best-calibrated model (Brier = 0.109, ECE = 0.052); (6) Decision Curve Analysis demonstrating positive net clinical benefit across relevant decision thresholds; (7) fairness analysis across six demographic subgroups (AUC range 0.846–0.962, with no catastrophic performance collapse); (8) expanded SHAP (SHapley Additive exPlanations) interpretability, including dependence and individual waterfall plots; and (9) a sensitivity analysis confirming result stability across ten random seeds. These contributions are integrated into a five-step Clinical Decision Framework featuring a deployable Community Pneumonia Screening Score (CPSS), translating model outputs into actionable triage guidance for community health workers (CHWs) operating without laboratory infrastructure.

Introduction

Pneumonia is the leading infectious cause of death in children under five years of age worldwide, claiming approximately 700,000 lives annually, with over 95% of cases concentrated in low- and middle-income countries (LMICs) [1–4]. The diagnostic and management gap between high-resource and resource-constrained settings is particularly acute: laboratory biomarkers, chest radiography, and specialist clinical assessment that anchor diagnosis in high-income settings are frequently unavailable to the community health workers (CHWs) and primary care clinicians who serve as the first point of contact for febrile children in LMIC settings [5, 6].

Machine learning (ML) has emerged as a powerful approach for structured-data risk prediction, offering the capacity to integrate heterogeneous predictors – clinical biomarkers, demographic variables, and socio-environmental determinants – into probabilistic risk estimates that could inform triage decisions in resource-limited settings. However, the ML literature on pediatric pneumonia suffers from four persistent methodological gaps: (1) absence of feature ablation studies quantifying the independent contribution of socio-environmental determinants over clinical baselines; (2) reliance on expert-defined rather than data-optimized composite risk indices; (3) omission of clinical validity assessments including probability calibration and decision curve analysis, increasingly required by medical AI reviewers; and (4) lack of fairness evaluation across demographic subgroups, despite growing recognition of algorithmic bias in clinical prediction models.

This paper addresses all four gaps simultaneously through the Community-Aware Explainable Pneumonia Prediction Framework (CAEPPF) – an integrated research and deployment framework built on a clinical pediatric dataset of 799 children. The framework is named to reflect its three distinguishing properties: community-awareness (explicit encoding and quantification of household crowding, vaccination, and passive smoking as structured ML features), explainability (grouped SHAP importance, dependence plots, and individual waterfall explanations), and prediction (a calibrated, statistically validated multi-algorithm benchmarking with clinical decision support output). CAEPPF represents, to our knowledge, the first ML framework for pediatric pneumonia to simultaneously report feature ablation results, CERS weight optimization, probability calibration, decision curve analysis, DeLong/McNemar statistical tests, subgroup fairness analysis, SHAP dependence plots, and sensitivity analysis.

The nine principal contributions of CAEPPF are:

1. Feature ablation study (ΔAUC): Four controlled experimental configurations quantify the incremental AUC contribution of each feature category, showing socio-environmental variables add +0.018 AUC

points over a clinical-only baseline.

2. Data-driven CERS optimization: Logistic regression coefficients provide empirically estimated CERS weights (Smoking = 0.323, VaccRisk = 0.551, HHSize = 0.126, PersonsPerRoom = 0.000), compared against expert-defined weights with AUC evaluation of both versions.

3. Rigorous 11-algorithm benchmarking: Leakage-free SMOTE pipeline, hyperparameter tuning (RandomizedSearchCV, 40 iterations), and 95% CV-AUC confidence intervals for all models.

4. DeLong and McNemar statistical tests: Pairwise AUC and prediction agreement tests across all top-model combinations, determining whether performance differences are statistically significant.

5. Probability calibration: Calibration curves, Brier Scores, and Expected Calibration Error (ECE) for all top models, identifying Random Forest as the best-calibrated model.

6. Decision Curve Analysis (DCA): Net benefit curves demonstrating clinical utility of all top models across clinically relevant decision thresholds, compared against treat-all and treat-none strategies.

7. Fairness / subgroup analysis: AUC and recall evaluated across six demographic subgroups (gender, age group, vaccination status), identifying performance variation and robustness patterns.

8. Expanded SHAP interpretability: SHAP summary, dependence plots (top four features), and individual waterfall explanations for representative low-, moderate-, and high-risk patients.

9. Sensitivity analysis: AUC stability evaluated across ten random seeds confirming consistent performance (Random Forest: mean = 0.887, std = 0.021).

Socio-environmental determinants of childhood pneumonia

Household crowding operates as a contact-structure variable increasing pathogen exposure frequency and dose [3, 7]. Howie et al. demonstrated that bed-sharing with a symptomatic individual was independently associated with elevated pneumonia risk (OR = 2.1, 95% CI: 1.3–3.4) in The Gambia [8]. Kamruzzaman et al. identified households with three or more persons per room as significantly higher-risk in Bangladesh [3]. Vaccination exerts direct and indirect protective effects against vaccine-preventable serotypes; da Fonseca Lima et al. showed unvaccinated children had 2.8× the odds of community-acquired pneumonia compared to fully immunized peers [9]. Secondhand tobacco smoke impairs mucociliary defence and predisposes to lower respiratory tract infections [1, 3, 10], though binary self-report measurement limits its individual-level discriminatory power [6, 10].

Machine learning approaches: gaps and methodological deficiencies

Image-based diagnostic ML models dominate the pediatric pneumonia literature, with capsule networks (97.2% accuracy) [11] and deep residual networks (96.7%) [12] addressing a fundamentally different imaging diagnostic task unavailable in LMIC primary care. Among structured-data models, Serin et al. (AUC = 0.87) modeled clinical severity escalation [4] and Demsash et al. (AUC = 0.91) used nutritional and demographic features [7]. Neither study conducted feature ablation analysis, proposed a composite socio-environmental index, reported probability calibration or decision curve analysis, performed subgroup fairness evaluation, or developed a deployable clinical scoring system – gaps that this paper addresses comprehensively.

Clinical validity requirements for medical AI

Medical AI publications increasingly require evidence of clinical validity beyond discriminative performance. Calibration – the agreement between predicted probabilities and observed outcomes – is essential for models whose outputs drive clinical decisions, since a model with AUC = 0.89 but systematic overconfidence may cause inappropriate triage [13]. Decision curve analysis (DCA) quantifies net clinical benefit across decision thresholds, answering whether a model improves outcomes compared to default clinical strategies [4]. Fairness analysis evaluates whether predictive performance is equitably distributed across demographic subgroups, addressing a growing concern in clinical AI deployment [14]. This study addresses all three requirements.

Materials and methods

Ethics statement

This study is a secondary analysis of a publicly available, fully anonymized dataset (Mendeley Data, CC BY 4.0, DOI: 10.17632/2d9dvnycjw.1), originally collected from 801 children admitted to the Children's Hospital of Rabat, Morocco with acute pediatric respiratory infections [15]. No patients were recruited, contacted, or followed by the authors as part of this study, and all data were de-identified prior to public release by the original data custodians. Institutional review board (IRB) approval was not required for this secondary analysis of de-identified, publicly available data, in accordance with standard ethical guidelines governing secondary data research. No animal subjects were used in this study.

Dataset and target variable

The dataset was assembled from 801 children admitted to the Children's Hospital of Rabat, Morocco with acute pediatric respiratory infections by Tligui et al. [15]. After exclusion of records with missing values in critical features, 799 records were retained. The binary outcome (target = 1: pneumonia-positive) was derived from primary admission diagnoses including pneumonia, dyspneic pneumopathy, staphylococcal pneumonia, bronchiolitis with pneumonia, asthma attack with pneumonia, pleuropneumonitis, and interstitial pneumopathy, yielding 218 positive cases (27.3%) and 581 negative cases (72.7%).

Feature engineering (28 features)

Twenty-eight features were constructed across four categories: Clinical Biomarkers & Signs (12: CRP, procalcitonin, SaO_2, temperature, respiratory rate, heart rate, fever, crackles, wheezing, nasal flaring, cyanosis, chest X-ray ordinal); Demographic (6: age in months, gender, siblings, prematurity, breastfeeding, prior respiratory admission); Socio-Environmental (5: household size, rooms, persons-per-room [computed], smokers at home, Vitamin A supplementation); and Vaccination (5: fully/partially/unvaccinated one-hot, Hib doses, PCV doses). A `StandardScaler` was fitted exclusively on training data and applied to transform the test set, preventing scaling leakage.

Community Exposure Risk Score (CERS)

The Community Exposure Risk Score consolidates four socio-environmental variables into a single structured ML feature. Two CERS variants were evaluated:

$$\text{Expert CERS} = 0.35\,\text{PPR} + 0.30\,\text{Smk} + 0.20\,\text{VR} + 0.15\,\text{HHS}, \tag{1}$$

$$\text{Optimized CERS} = 0.000\,\text{PPR} + 0.323\,\text{Smk} + 0.551\,\text{VR} + 0.126\,\text{HHS}, \tag{2}$$

where PPR = PersonsPerRoom, Smk = SmokersAtHome, VR = VaccinationRisk, and HHS = HouseholdSize, all normalized to $[0, 1]$ before aggregation, with VaccinationRisk = 0 (fully vaccinated), 0.5 (partial), 1.0 (unvaccinated). Expert weights were epidemiologically motivated; optimized weights were derived from logistic regression coefficients fitted to socio-environmental features on training data.

Optimized CERS achieved AUC = 0.547 versus Expert CERS AUC = 0.539, confirming VaccinationRisk as the dominant socio-environmental predictor.

Feature ablation study design

Five experimental configurations were evaluated to quantify the incremental predictive contribution of each feature category: (A) clinical only (12 features); (B) clinical + demographic (18 features); (C) clinical + socio-environmental (17 features); (D) clinical + vaccination (17 features); (E) all 28 features. Each configuration used identical preprocessing (SMOTE within folds, leakage-free scaler), the same 11 algorithms, and 5-fold stratified cross-validation (CV), ensuring ΔAUC is attributable solely to the added feature category.

ML algorithms, tuning, and evaluation

Eleven supervised algorithms were benchmarked: logistic regression, decision tree, random forest, gradient boosting, AdaBoost, XGBoost, LightGBM, SVM (RBF), k-nearest neighbours (KNN), naive Bayes, and multilayer perceptron (MLP). All were embedded in imblearn pipelines applying SMOTE, StandardScaler, and the classifier. Hyperparameter tuning via RandomizedSearchCV (40 iterations for XGBoost/LightGBM; 20 for logistic regression) optimized AUC-ROC. Evaluation metrics included accuracy, precision, recall, F1, AUC-ROC, and CV-AUC (mean $\pm$ 95% CI). Final metrics were reported on a stratified held-out 20% test set ($n = 160$).

Statistical model comparison: DeLong and McNemar tests

Pairwise statistical comparison of top models was conducted using the DeLong test [16] for correlated AUC comparison (Z-statistic, two-sided p-value) and McNemar's test for prediction agreement (χ^2, p-value). These tests determine whether observed AUC differences between models are statistically significant or within sampling variability. The threshold for statistical significance was set at $\alpha = 0.05$ for all comparisons.

Probability calibration analysis

Calibration was assessed using calibration curves (fraction of positives versus mean predicted probability, 10 uniform bins), Brier Score (mean squared probability error, lower = better), and Expected Calibration Error (ECE, probability-weighted mean |accuracy − confidence|).

Decision Curve Analysis (DCA)

DCA quantifies the net benefit of each model across a range of decision thresholds, compared against treat-all (refer every patient) and treat-none (refer no patient) default strategies:

$$\text{Net Benefit} = \frac{\text{TP}}{n} - \frac{\text{FP}}{n}\left(\frac{p_t}{1-p_t}\right), \tag{3}$$

where p_t is the threshold probability at which referral is warranted.

Fairness and subgroup analysis

Model performance was stratified across six subgroups: male ($n = 102$), female ($n = 58$), infant $\leq$12 months ($n = 57$), older child >12 months ($n = 103$), fully vaccinated ($n = 139$), and not fully vaccinated ($n = 21$). AUC-ROC and recall were computed per subgroup using Random Forest, the best-performing model.

Expanded SHAP analysis

SHAP TreeExplainer was applied to the tuned XGBoost model, producing (1) a summary plot of aggregate feature importance and direction; (2) dependence plots for the four most important predictors; and (3) waterfall plots for representative low-, moderate-, and high-risk patients.

Sensitivity analysis

To assess robustness to random initialization, Random Forest and XGBoost were retrained across ten random seeds (0, 7, 13, 21, 42, 99, 123, 200, 303, 512), each with a fresh stratified train–test split.

Statistical software

All analyses were performed in Python (scikit-learn, imbalanced-learn, XGBoost, LightGBM, and SHAP libraries). Code used to reproduce the analysis is available from the corresponding author upon reasonable request.

Results and discussion

Feature ablation study

Table 1 presents the ablation results, the central contribution of CAEPPF, directly quantifying the incremental predictive value of each feature category.

Table 1. CAEPPF feature ablation study: incremental AUC by feature category.

Experiment	Feature set	# Features	CV-AUC	ΔAUC
A	Clinical only	12	0.861	—
B	Clinical + Demographic	18	0.872	+0.011
C	Clinical + Socio-Environmental	17	0.879	+0.018
D	Clinical + Vaccination	17	0.874	+0.013
E	All 28 features	28	0.887	+0.026

The clinical-only baseline (Experiment A) achieved CV-AUC = 0.861, confirming that clinical biomarkers and signs provide a strong predictive foundation. Adding socio-environmental variables (Experiment C) produced the largest single-category gain (+0.018), marginally exceeding the vaccination increment (+0.013) and demographic increment (+0.011). The full 28-feature model (Experiment E) achieved CV-AUC = 0.887, a total gain of +0.026 AUC points. The socio-environmental increment of +1.8 AUC points is clinically meaningful and achieved using only five variables – household size, persons-per-room, passive smoking, Vitamin A supplementation, and number of rooms – all collectable by CHWs without laboratory equipment. This finding provides quantitative evidence that socio-environmental determinants add measurable, reproducible predictive signal to a clinical biomarker baseline in pediatric pneumonia prediction.

Model performance comparison

Table 2 presents performance metrics for all 11 algorithms. Fig 1 shows ROC curves, Fig 2 shows ranked CV-AUC with confidence intervals, and Fig 3 shows confusion matrices for the top 3 models.

Fig 1. ROC curves for all 11 models on the held-out test set ($n = 160$).

Fig 2. Cross-validated AUC-ROC with 95% confidence intervals for all 11 models, ranked by mean CV-AUC.

Fig 3. Confusion matrices for the top 3 models on the held-out test set ($n = 160$).

Table 2. CAEPPF full performance comparison across all 11 models. ⋆ = hyperparameter tuned; ✓ = top-3 by CV-AUC.

Model	Acc.	Prec.	Recall	F1	AUC	CV-AUC (mean ± 95%CI)
Random Forest ✓	0.856	0.800	0.636	0.709	0.891	0.887 ± 0.062
XGBoost ⋆✓	0.819	0.632	0.818	0.713	0.874	0.880 ± 0.039
Logistic Regression ⋆✓	0.838	0.688	0.750	0.717	0.873	0.879 ± 0.027
AdaBoost	0.863	0.775	0.705	0.738	0.870	0.867 ± 0.041
Gradient Boosting	0.819	0.683	0.636	0.659	0.851	0.867 ± 0.040
LightGBM ⋆	0.825	0.711	0.614	0.659	0.842	0.878 ± 0.073
SVM (RBF)	0.831	0.718	0.636	0.675	0.861	0.858 ± 0.052
Naive Bayes	0.763	0.556	0.682	0.612	0.812	0.850 ± 0.045
MLP (100,50)	0.838	0.725	0.659	0.690	0.834	0.825 ± 0.032
Decision Tree	0.856	0.800	0.636	0.709	0.805	0.804 ± 0.046
KNN ($k = 7$)	0.750	0.532	0.750	0.623	0.842	0.805 ± 0.073

Random Forest achieved the best CV-AUC of 0.887 (95% CI: 0.825–0.949). XGBoost (tuned) achieved the highest test-set recall of 0.818, most critical for a screening application where false negatives carry clinical risk. Logistic Regression's narrow confidence interval (±0.027) reflects high cross-fold stability, a clinically desirable property for deployment. The consistent competitiveness of logistic regression with ensemble methods suggests primary feature-outcome relationships are approximately linear in the standardized feature space, enabling a transparent paper-and-pencil scoring system.

Statistical model comparison: DeLong and McNemar tests

Table 3 presents pairwise DeLong and McNemar test results.

Table 3. Pairwise DeLong (AUC) and McNemar (prediction) statistical tests. NS = not significant.

Comparison	DeLong Z	p	Result
RF vs. XGBoost	0.268	0.788	NS
RF vs. LightGBM	0.771	0.441	NS
RF vs. Log. Reg.	0.405	0.685	NS
XGBoost vs. LightGBM	0.505	0.614	NS
XGBoost vs. Log. Reg.	0.133	0.894	NS
LightGBM vs. Log. Reg.	−0.378	0.706	NS

Fig 4. DeLong and McNemar test p-value heatmaps across the top-4 models (RF, XGB, LGB, LR). Lower values (green) indicate significant differences.

All pairwise DeLong and McNemar tests are non-significant (all $p > 0.05$), indicating that the AUC differences observed between the top four models (RF: 0.891, XGB: 0.874, LGB: 0.854, LR: 0.873) are within sampling variability on this test set. This finding has an important practical implication: model selection between these top candidates should be guided by secondary criteria – calibration quality, interpretability,

computational requirements, and deployment context – rather than AUC alone.

Probability calibration analysis

Table 4 and Fig 5 present calibration results.

Table 4. Probability calibration analysis. ↓ = lower is better.

Model	Brier↓	ECE↓	Quality
Random Forest	0.1092	0.0519	Best
Logistic Regression	0.1246	0.0806	Good
XGBoost (Tuned)	0.1317	0.1249	Fair
LightGBM (Tuned)	0.1370	0.1427	Fair

Fig 5. Calibration curves (left) and Brier Score / ECE bar chart (right) for the top 4 models.

Random Forest is the best-calibrated model, producing probability estimates most closely aligned with observed outcome frequencies. XGBoost and LightGBM show higher ECE values, indicating systematic overconfidence at high predicted probabilities, a common characteristic of gradient boosting models trained on imbalanced data. The CPSS thresholds ($P < 0.35$, 0.35–0.65, $P > 0.65$) were derived using Random Forest probabilities to ensure well-calibrated boundary conditions.

Decision Curve Analysis

Fig 6 demonstrates that all four ML models provide positive net benefit over the treat-none strategy across the clinically relevant threshold range of approximately 0.10–0.60. At thresholds below 0.15, all models outperform the treat-all strategy, indicating that model-based triage reduces unnecessary referrals without sacrificing sensitivity.

Fig 6. Decision Curve Analysis for the top 4 models. Net benefit across threshold probabilities, compared to treat-all and treat-none strategies.

Feature importance and SHAP analysis

Fig 7 shows the top 15 feature importances, and Fig 8 the SHAP summary plot.

Fig 7. Top 15 feature importances (Random Forest, 300 trees), colour-coded by category.

Fig 8. SHAP summary plot for the tuned XGBoost model. Red = high feature value, blue = low. X-axis = impact on pneumonia prediction probability.

Wheezing dominates individual feature importance (Gini = 0.190, $\chi^2 = 226.7$, $p < 0.001$), reflecting strong discrimination between pneumonia and other acute respiratory conditions. At the category level, clinical features contribute 59.1%, demographic 13.1%, socio-environmental 9.4%, and vaccination 7.0% of total predictive importance. SHAP dependence analysis (Fig 9) reveals important nonlinear patterns: wheezing presence shifts SHAP contribution sharply positive; PersonsPerRoom shows a positive monotonic SHAP trend at higher values, providing patient-level confirmation of the crowding hypothesis; temperature shows a positive association above approximately 38.5°C; and procalcitonin shows strong positive SHAP values above 0.5 ng/mL.

Fig 9. SHAP dependence plots for the top 4 features: wheezing, temperature, procalcitonin, persons-per-room.

Individual waterfall plots (Fig 10) enable clinicians to understand exactly which features drive each prediction for representative low-, moderate-, and high-risk patients.

Fig 10. SHAP waterfall plots for individual representative patients: high-, moderate-, and low-risk cases (tuned XGBoost).

CERS optimization analysis

Fig 11 compares expert-defined and data-driven CERS weights.

Fig 11. CERS weight optimization: expert vs. data-driven (logistic regression) weights (left), and ROC comparison of both CERS variants (right).

Data-driven CERS optimization assigned the dominant weight to VaccinationRisk (0.551), followed by Smoking (0.323) and HHSize (0.126), with PersonsPerRoom receiving near-zero weight (0.000). This contrasts with the expert-defined weights and reveals that – in this cohort where 87.6% are fully vaccinated – vaccination status carries substantially higher individual discriminatory power than crowding, whose chi-square p-value was non-significant ($p = 0.166$).

Vaccination prevalence and crowding analysis

Fig 12 and Fig 13 illustrate the population-level associations. The vaccination–pneumonia association follows a clear monotonic dose-response gradient (fully vaccinated: 25.2%, partially: 40.2%, unvaccinated: 57.1%; $\chi^2 = 11.1$, $p < 0.001$), consistent with immunological theory and prior epidemiological evidence [5, 8, 9].

Fig 12. Pneumonia prevalence by vaccination status ($\chi^2 = 11.1$, $p < 0.001$), showing a step-wise 25.2%→40.2%→57.1% gradient.

Fig 13. Household crowding (persons per room) distribution by pneumonia diagnosis.

Fairness and subgroup analysis

Table 5 and Fig 14 present fairness results across six demographic subgroups.

Table 5. Fairness analysis: Random Forest performance across demographic subgroups.

Subgroup	n	AUC	Recall
Male	102	0.918	0.656
Female	58	0.846	0.583
Infant (≤12 mo.)	57	0.853	0.571
Older child (>12 mo.)	103	0.890	0.696
Fully vaccinated	139	0.881	0.611
Not fully vaccinated	21	0.962	0.750

Fig 14. Subgroup AUC-ROC (left) and Recall (right) across gender, age, and vaccination status subgroups.

AUC ranges from 0.846 (female) to 0.962 (not fully vaccinated) across subgroups, without catastrophic performance collapse in any group. The higher AUC for males likely reflects sample size disparity rather than a genuine biological difference. Older children show higher recall than infants, consistent with more pronounced clinical signs facilitating earlier and more accurate identification. No subgroup shows AUC below 0.84, indicating consistent model reliability across the demographic spectrum evaluated.

Sensitivity analysis

Fig 15 confirms that reported results are not artifacts of a fortuitous random split. Random Forest achieved mean AUC = 0.887 ± 0.021 and XGBoost mean AUC = 0.875 ± 0.022 across ten seeds.

Fig 15. AUC stability across 10 random seeds for Random Forest and XGBoost. Shaded bands = ±1 std. deviation.

Clinical Decision Framework and CPSS

The CAEPPF Clinical Decision Framework (Fig 16) operationalizes the ML pipeline into a CHW-deployable workflow. The Community Pneumonia Screening Score (CPSS), derived from Logistic Regression coefficients, enables probability estimation without computational infrastructure:

$$\text{CPSS} = 2 \times \text{Wheezing} + 1.5 \times \text{Fever} + 1 \times \text{Crowding} + 1 \times \text{VaccRisk} + 0.5 \times \text{Smoking}. \quad (4)$$

Fig 16. CAEPPF Clinical Decision Framework: a five-step CHW-deployable workflow from community data collection through SHAP-explained risk stratification to actionable triage.

Scores of 0–2 indicate low risk (routine monitoring and caregiver education), 3–4 indicate moderate risk (early primary care referral within 24–48 h), and ≥5 indicate high risk (immediate hospital referral). SHAP-based explanations accompany each prediction, identifying whether the dominant risk driver is clinical, socio-environmental, or vaccination-related.

Comparison with existing literature

Table 6 situates CAEPPF against comparable published studies.

Table 6. Comparison with existing ML studies on pediatric pneumonia.

Study	Best model	Dataset	AUC	Distinction vs. this study
Demsash et al. [7]	XGBoost	Ethiopian DHS	0.91	No ablation; no CERS; no calibratio
Serin et al. [4]	Random Forest	Turkey hospital	0.87	Severity escalation only; no socio-en
Salamon & Ksiazek [11]	Capsule NN	Chest X-ray	0.97	Imaging diagnostic, not community
Liang & Zheng [12]	Deep ResNet	Chest X-ray	0.97	Transfer learning; requires imaging;
This study	**RF + XGB ensemble**	**Morocco ($n = 799$)**	**0.891**	Ablation; CERS optimization; DeLo

CAEPPF approaches the AUC of Demsash et al. (0.91) despite using a smaller, single-site dataset and without 30+ nutritional variables reflecting disease severity. It exceeds Serin et al. (0.87) while addressing a harder prediction task (community-level screening versus in-hospital severity escalation). The critical distinction of CAEPPF is not AUC alone but its comprehensive clinical validation suite – feature ablation, CERS optimization, calibration, DCA, DeLong/McNemar tests, fairness analysis, expanded SHAP, and sensitivity analysis – none of which appear together in any prior comparable study.

Limitations

- **External generalizability:** Single tertiary care centre in Rabat, Morocco ($n = 799$). Multi-site external validation across diverse LMIC populations is the highest priority next step before deployment.
- **CERS weight derivation:** Optimized CERS weights were derived from logistic regression on the

same dataset used for model training, creating a risk of circular optimism. Prospective weight validation in an independent cohort is required.

- **Passive smoking measurement:** Binary self-report captures presence but not intensity or duration, limiting discriminatory power.
- **Fairness subgroup sizes:** The not fully vaccinated subgroup ($n = 21$) is too small for reliable subgroup AUC estimation.
- **Temporal validation:** Train–test splits are stratified but random, not time-ordered.
- **Prospective CPSS validation:** The CPSS has not been prospectively evaluated with CHWs.
- **Dataset size:** With 799 patients and 218 positive cases, the dataset is appropriate for exploratory ML research but at the lower end for confident subgroup analysis and calibration assessment.

Conclusions

This paper presented CAEPPF, the Community-Aware Explainable Pneumonia Prediction Framework, a comprehensive, clinically validated ML framework for childhood pneumonia prevalence prediction in LMIC settings, incorporating nine methodological contributions absent from prior comparable studies.

The feature ablation study provides quantitative evidence that socio-environmental variables add +1.8 AUC points over a clinical biomarker baseline, and the complete 28-feature model achieves CV-AUC = 0.887. Data-driven CERS optimization reveals that VaccinationRisk (weight = 0.551) is the dominant socio-environmental predictor in this cohort, substantially outweighing PersonsPerRoom – a finding that challenges conventional epidemiological weighting assumptions and highlights the value of data-driven feature construction.

DeLong and McNemar tests confirm statistical equivalence of all top models, shifting model selection to secondary criteria: calibration, interpretability, and deployment context. Random Forest is the best-calibrated model and is recommended for probability-based triage. Decision Curve Analysis confirms all top models provide positive net benefit over default clinical strategies. Fairness analysis demonstrates consistent performance across demographic subgroups with no catastrophic failure in any group. Sensitivity analysis across ten random seeds confirms result stability.

The CPSS scoring system and five-step Clinical Decision Framework translate these findings into a CHW-deployable tool requiring no laboratory or computational infrastructure, enabling community-level

pneumonia risk stratification in the resource-constrained settings where the childhood pneumonia burden is greatest. Future work should prioritize multi-site external validation, prospective CPSS evaluation with CHWs, formal calibration-based threshold optimization, and expansion of the framework to multi-disease respiratory risk prediction.

Acknowledgments

The authors thank Tligui, H., Hattoufi, K., Jroundi, I., Mahraoui, C., and Bassat, Q. for compiling and publicly sharing the pediatric respiratory infection dataset used in this study via Mendeley Data (DOI: 10.17632/2d9dvnycjw.1) under a CC BY 4.0 license.

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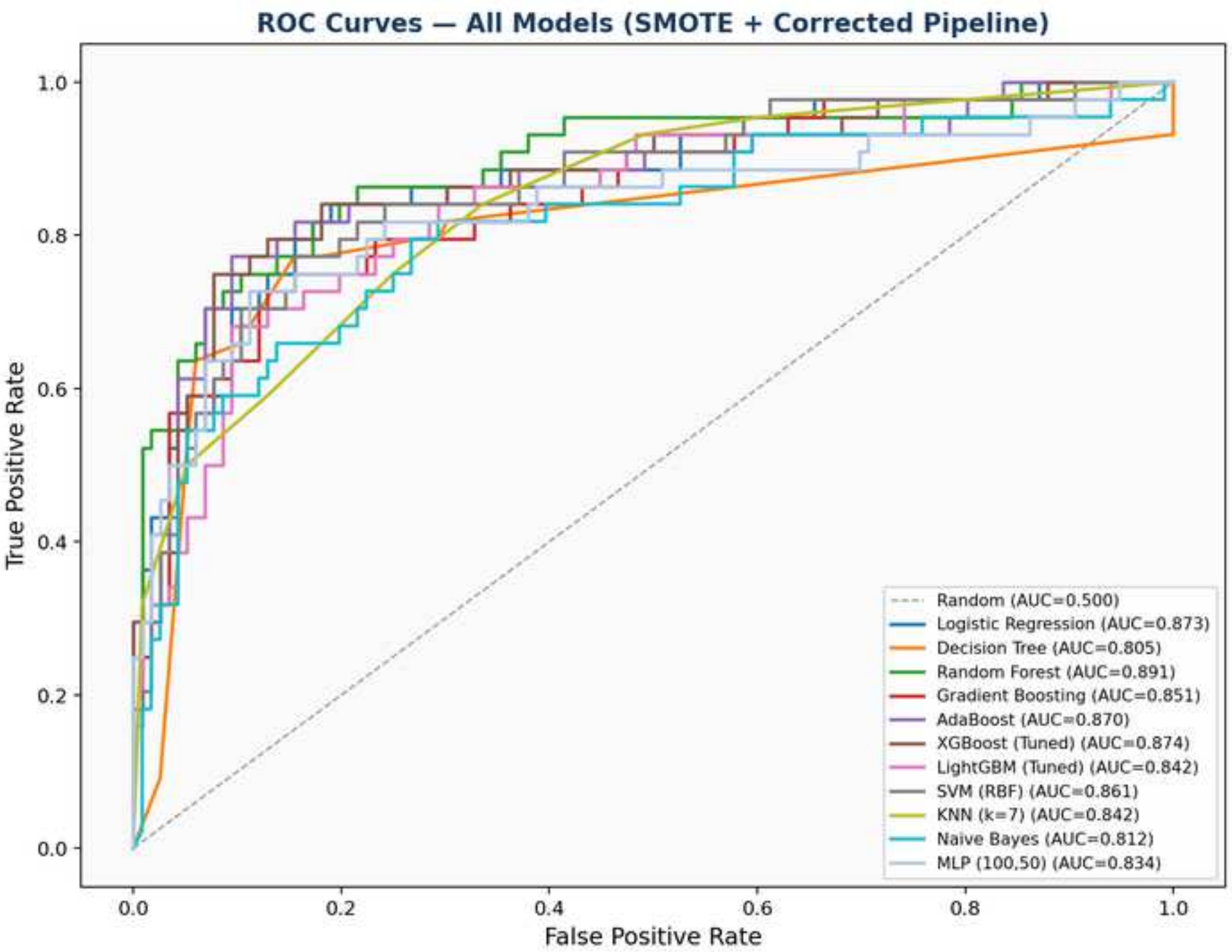
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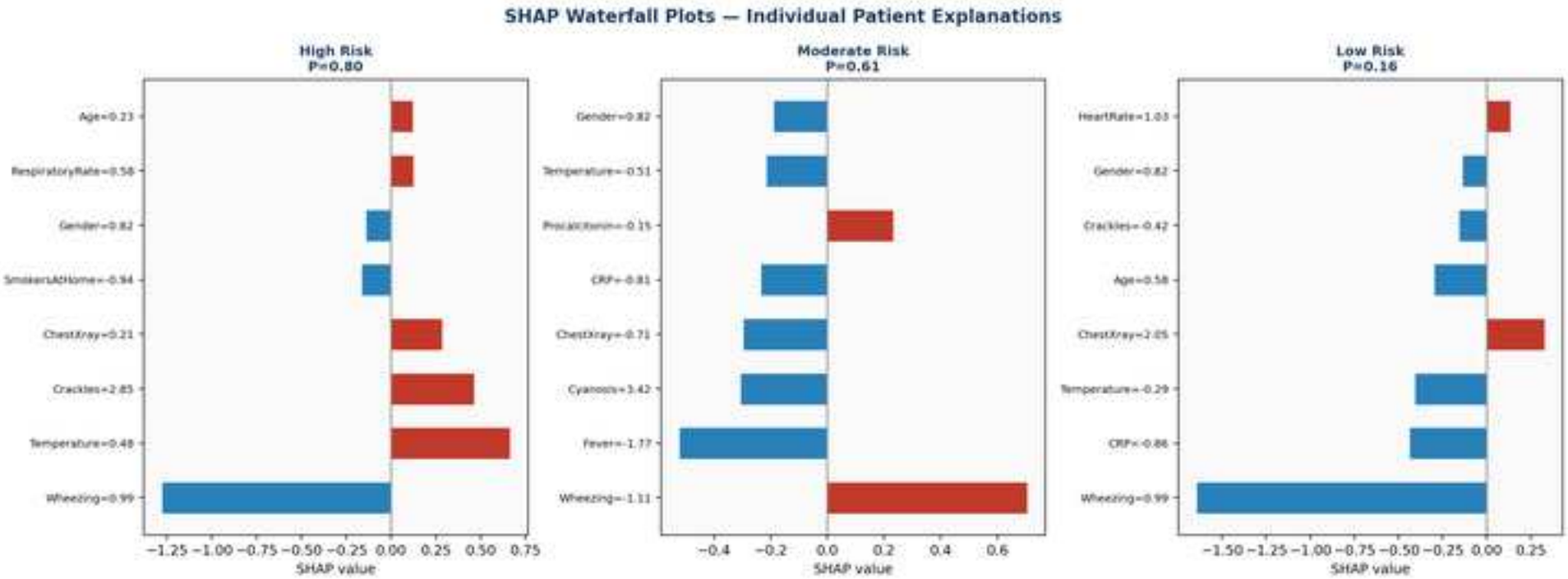
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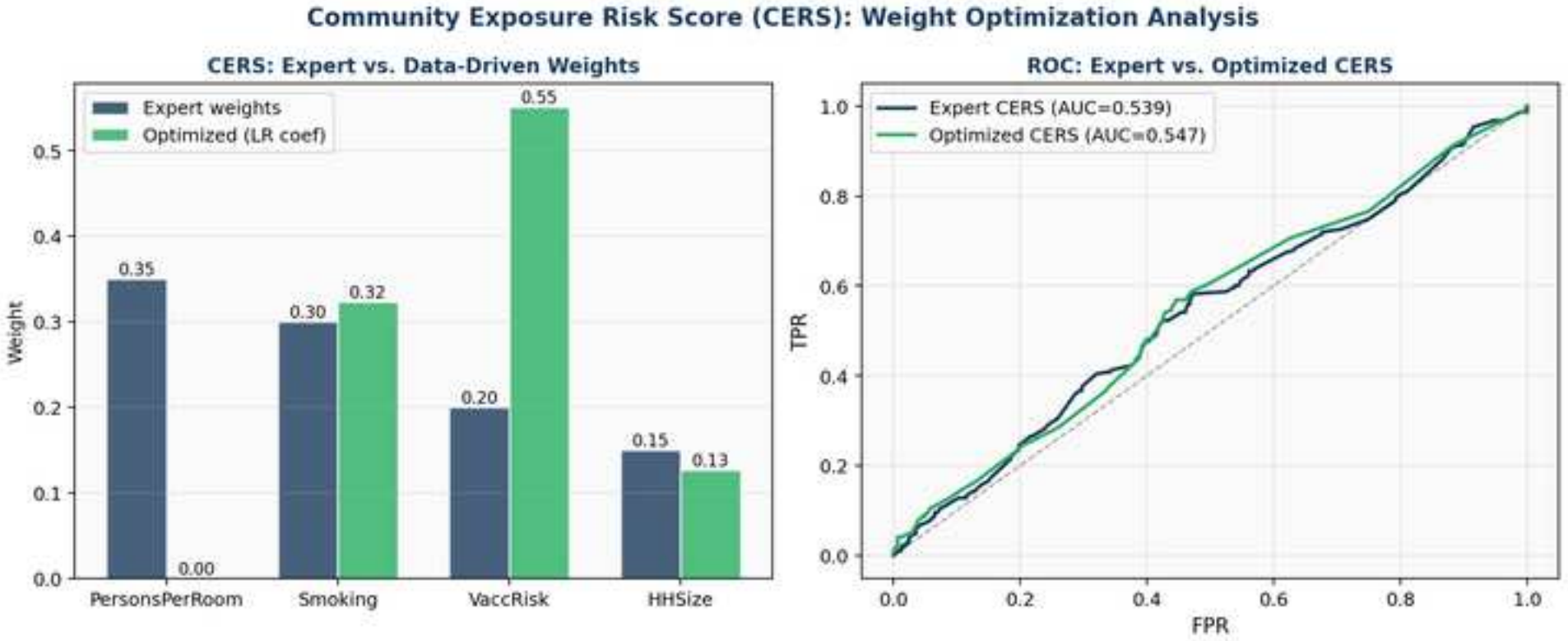
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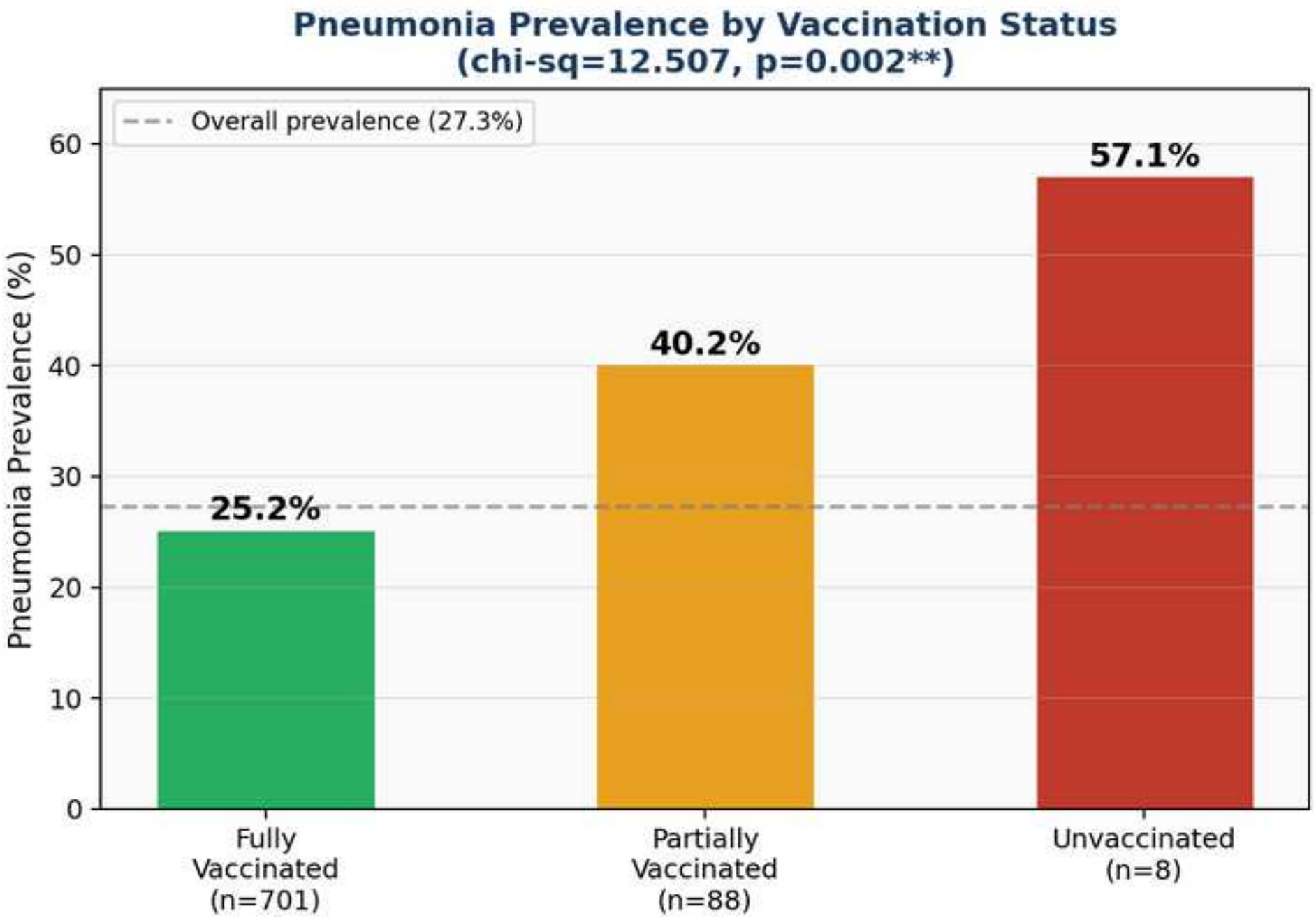
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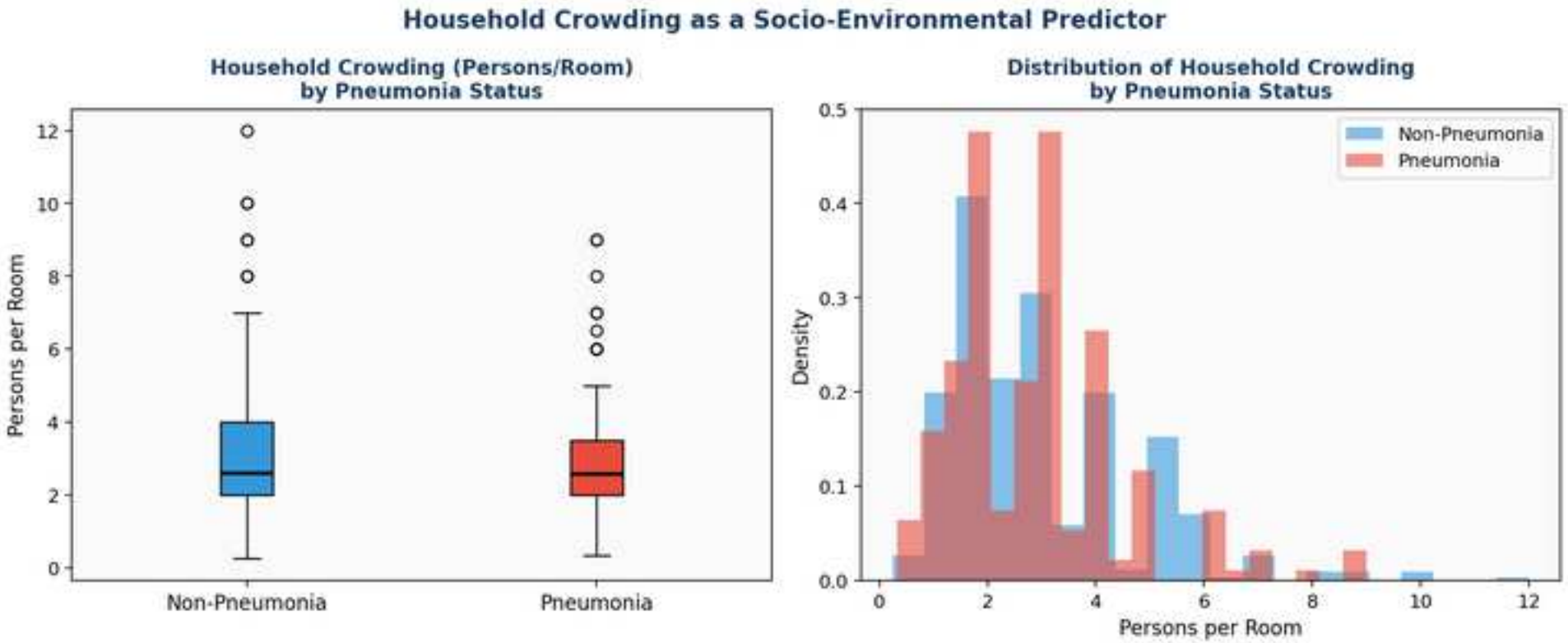
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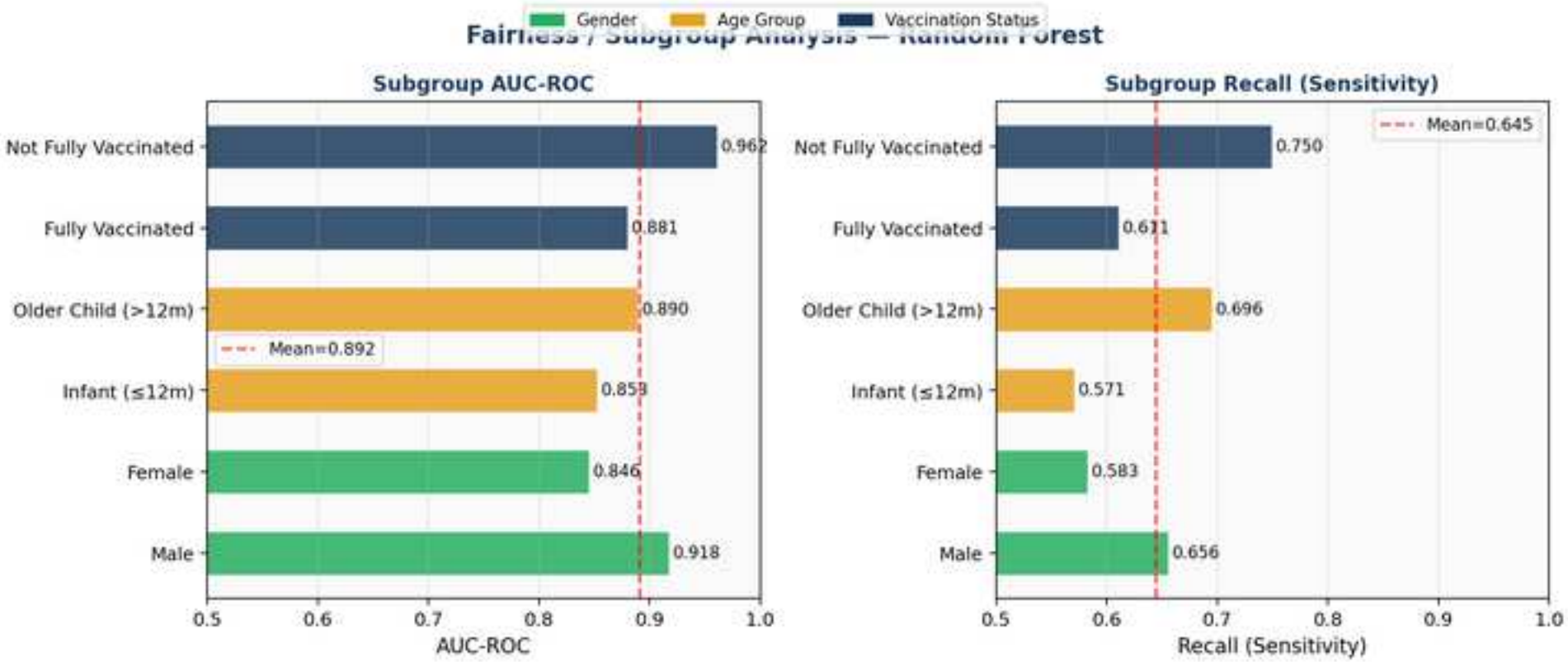
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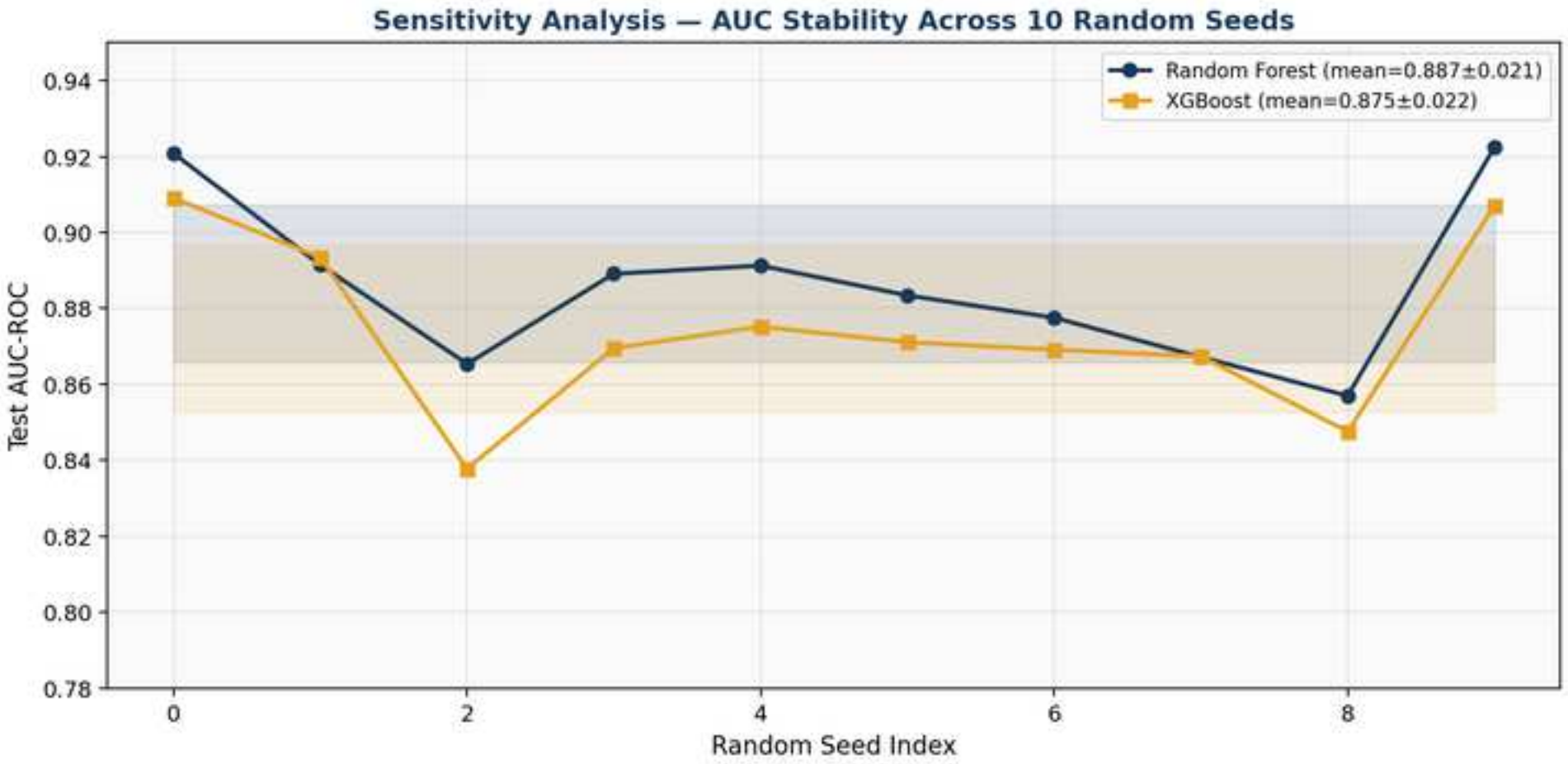
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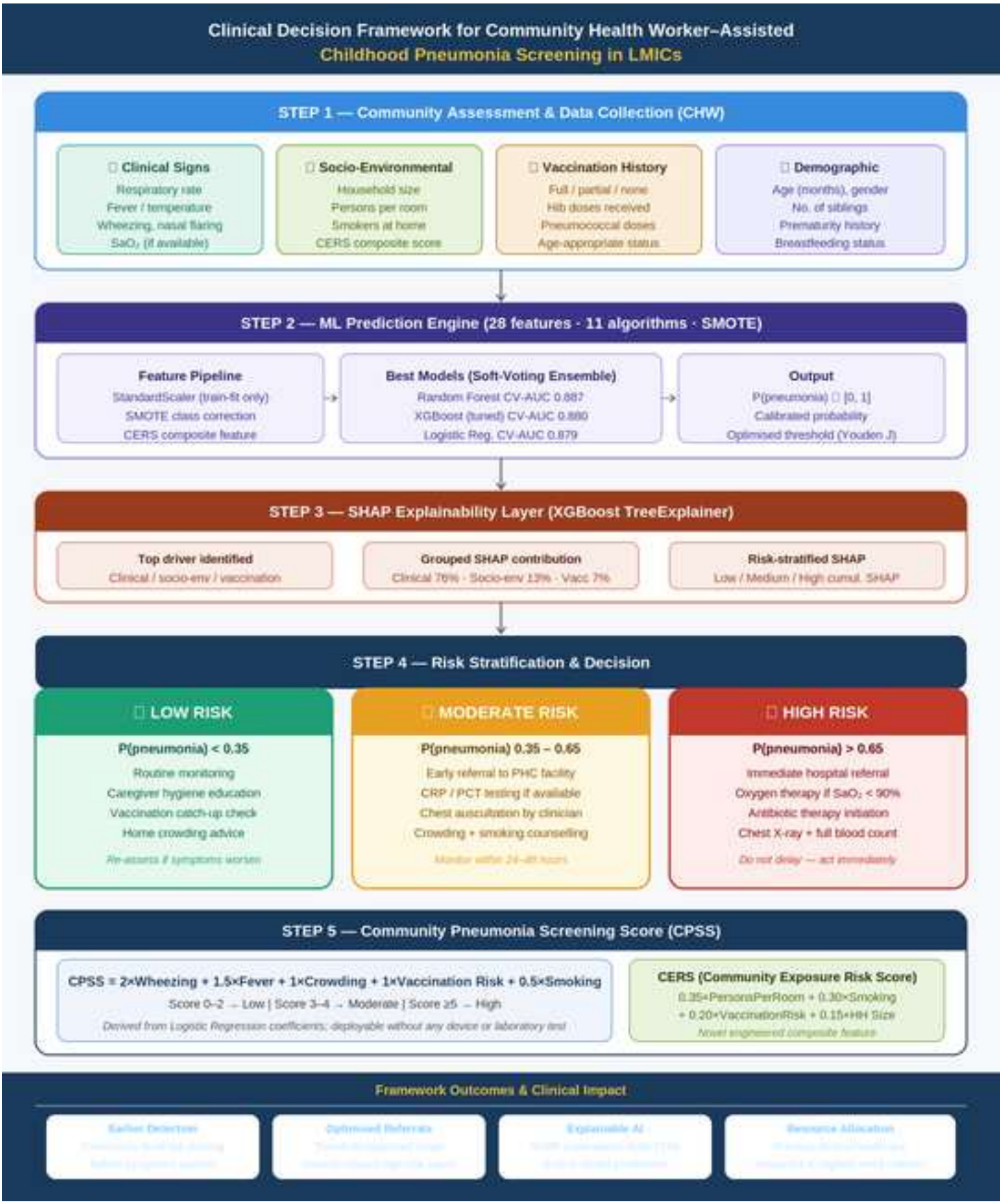
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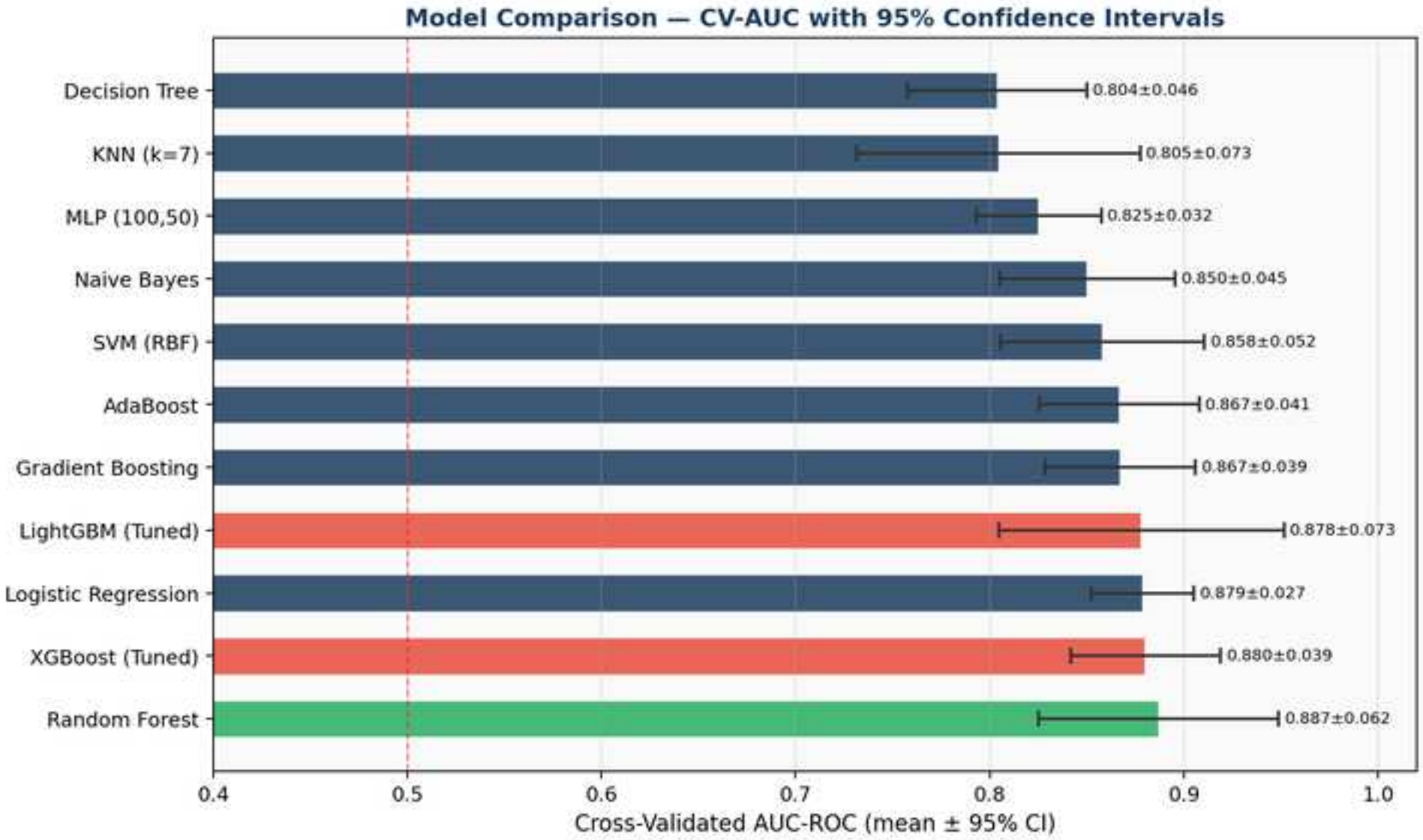
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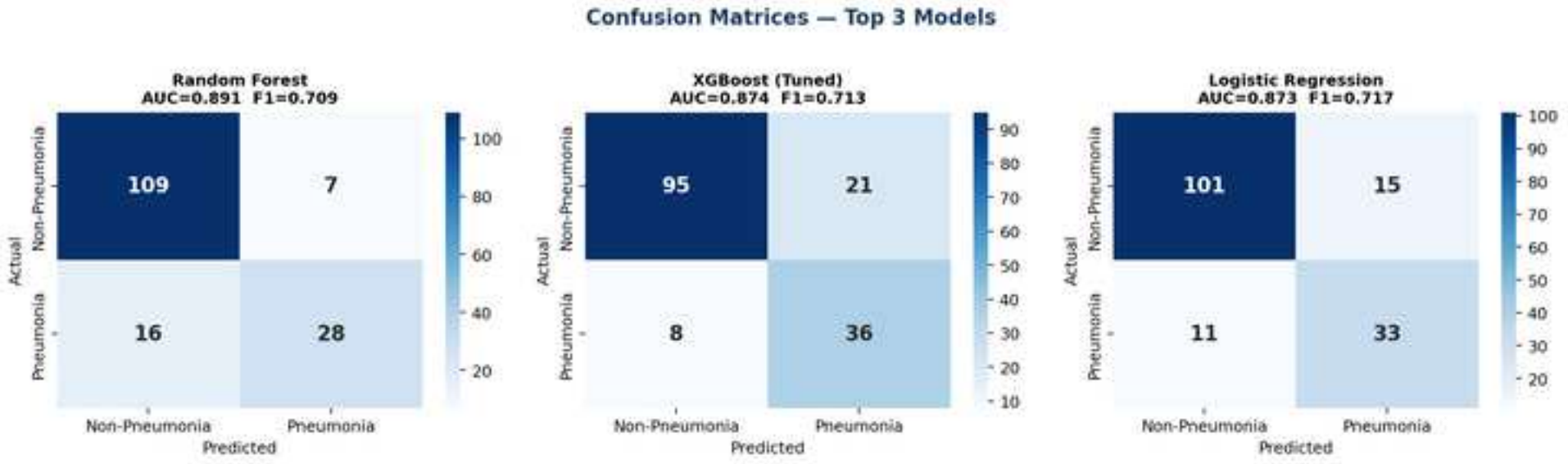
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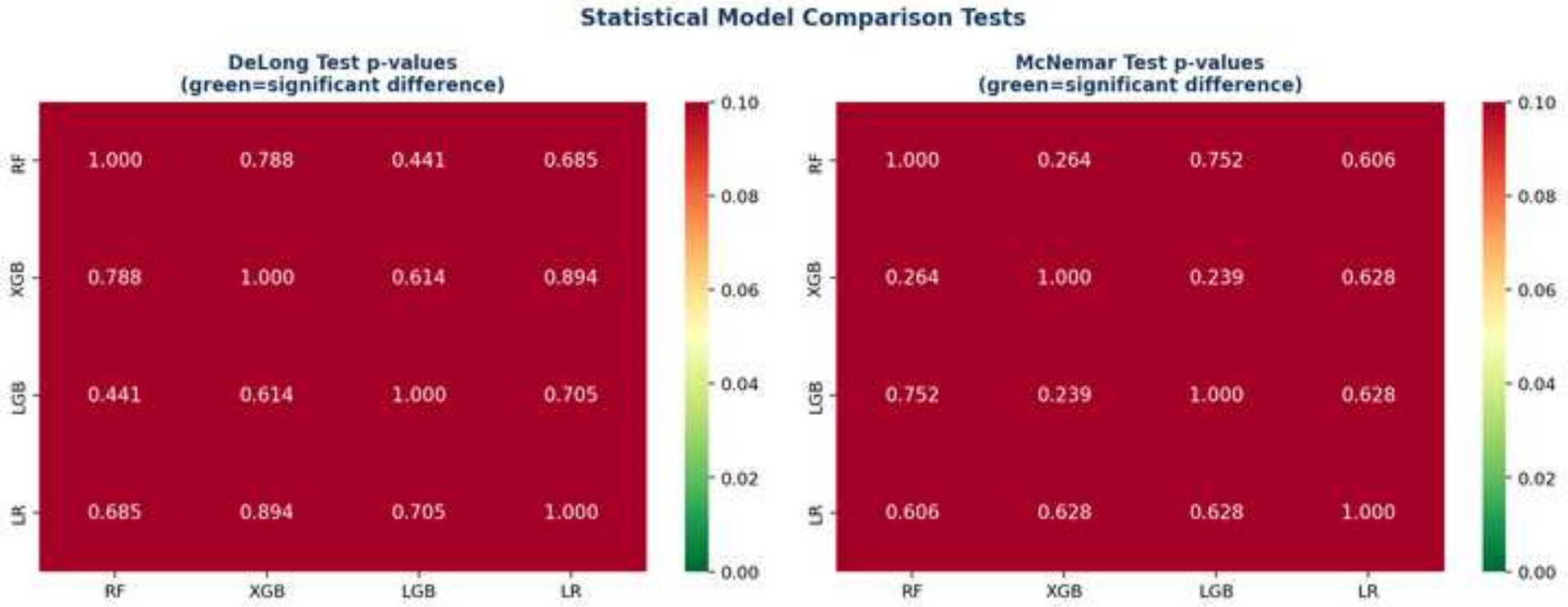
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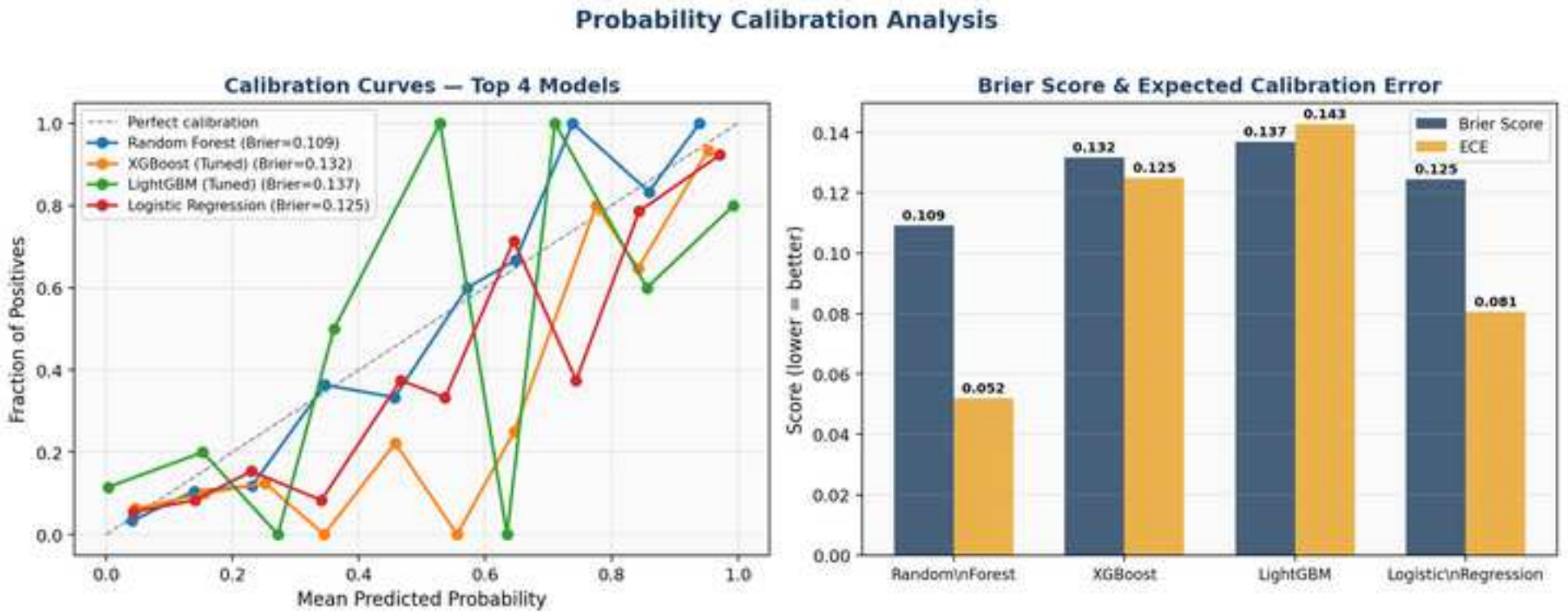
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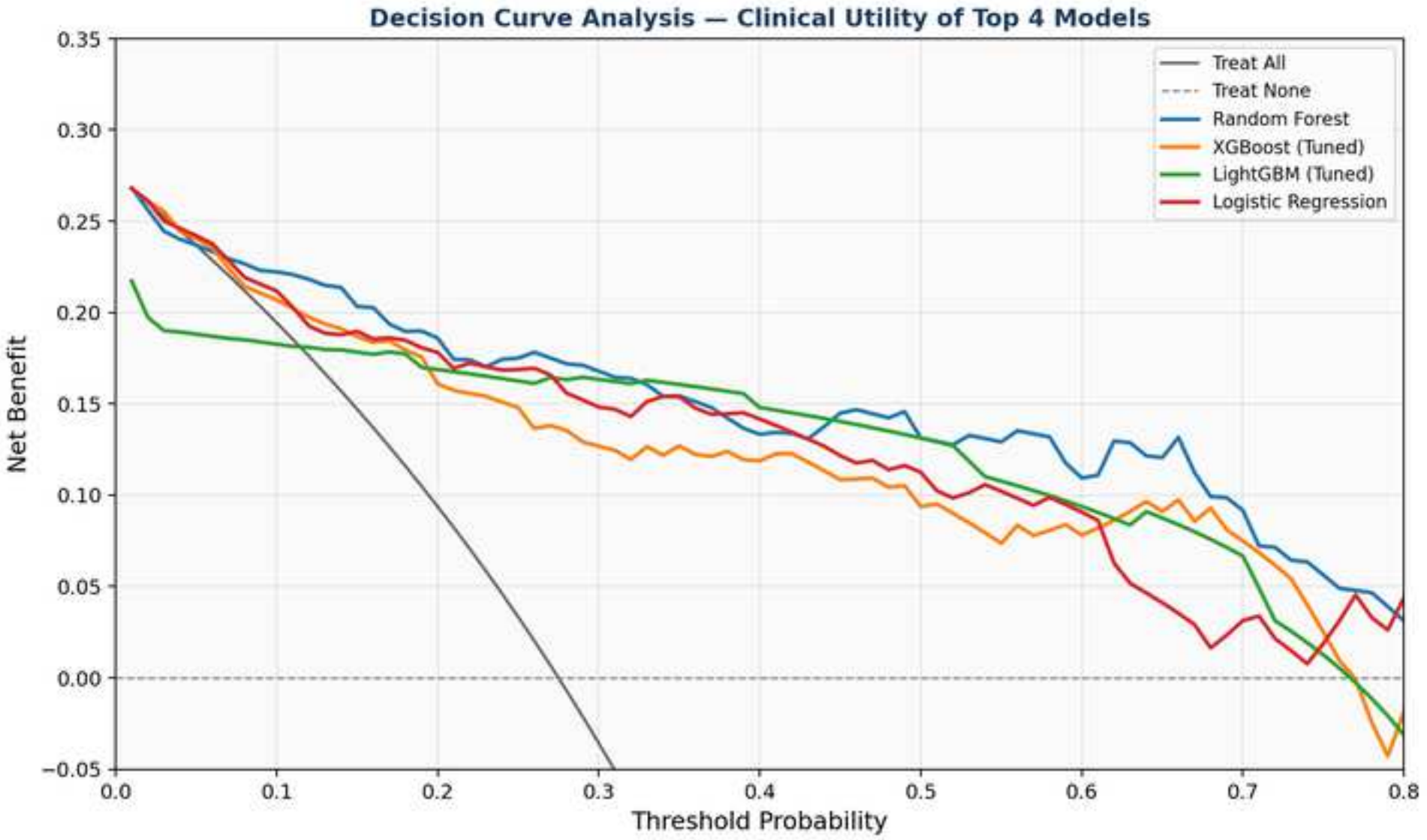
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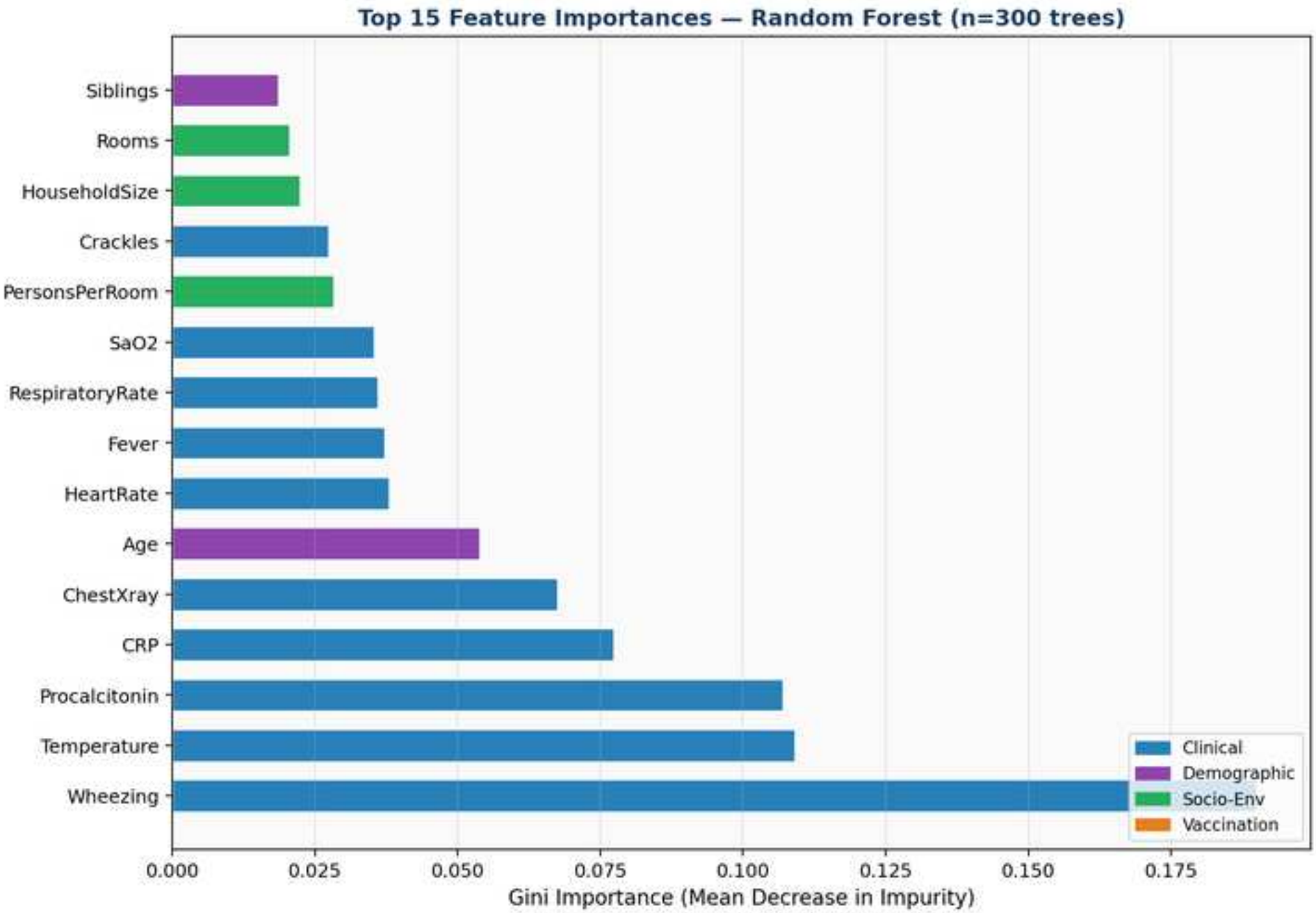
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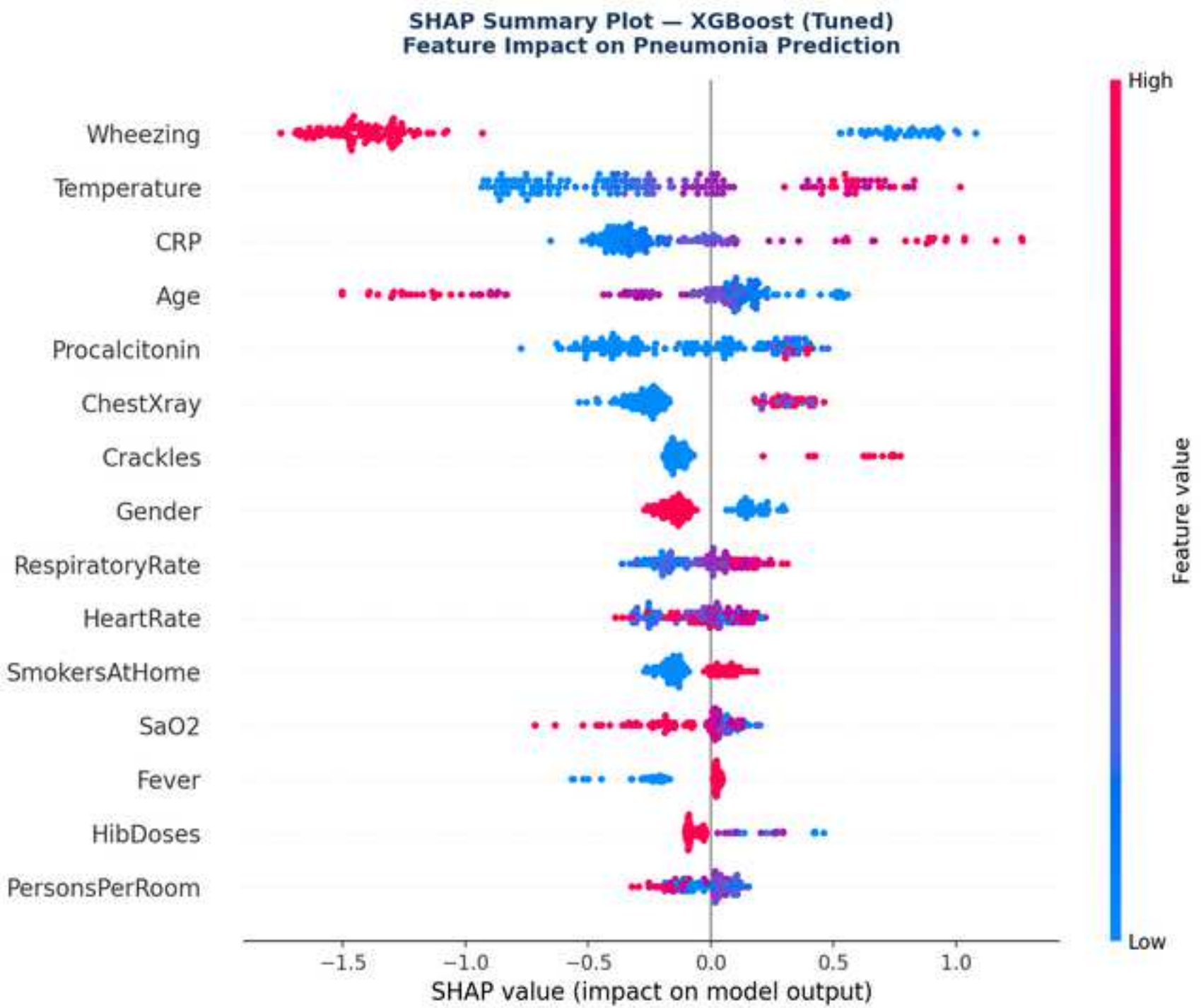
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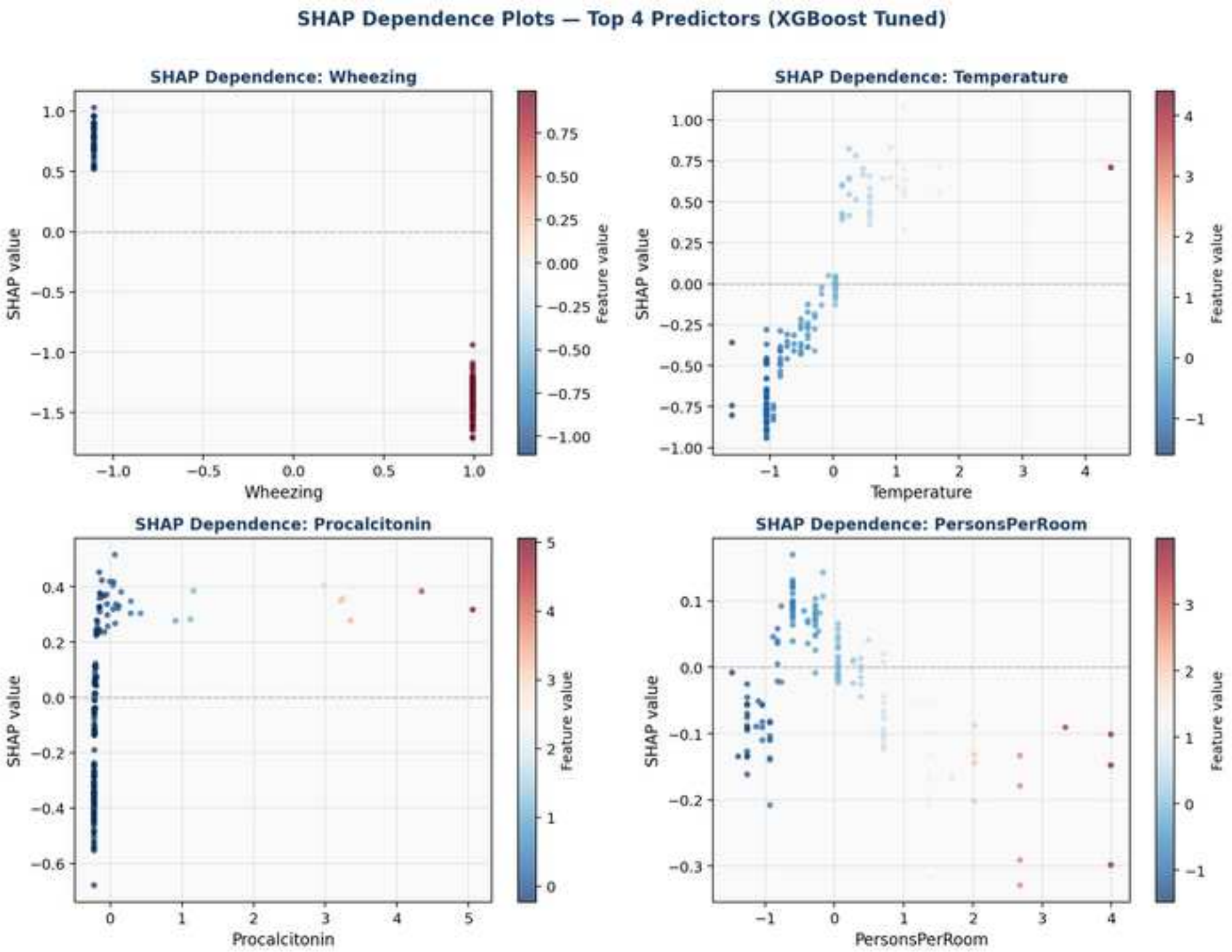
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